

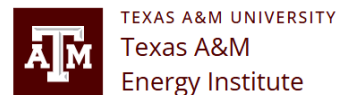
Repurposing Stranded Natural Gas: A Technical and Economic Analysis of Modern Monetization Pathways

Digital Pipelines & Refineries, and Modularized Midstream Plants

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Executive Summary

The current global energy landscape is undergoing an unprecedented period of transformation, driven by both breakthroughs in new energy technologies, as well as demand for high-density compute and industrial power. This report includes a comparative techno-economic analysis of estimated annual revenue streams for stranded natural gas from three different types of processes: physical pipelines, modular compute, and modular on-site conversion, and explores the advantages and disadvantages of each approach. While the main focus is on the newer digital monetization pathways of bitcoin and AI compute, the other options are included for a more comprehensive comparison.

The potential supply of global stranded natural gas fuel is enormous. As of 2023, there is an estimated 148 billion cubic meters of gas flaring globally, and even that may be an understatement.¹ In addition to the massive amount of energy waste that results from this flaring, there is of course the environmental issue of methane released into the atmosphere. By capturing this stranded gas and using it for generation or another midstream process, all of these monetization pathways can be effectively viewed as “methane scrubbers” for the energy industry. Flaring alone has been demonstrated to be inefficient, with one report finding that flares are only effective in removing 91% of methane versus 99% for certain on-site generators.² This reduction is significant, and important, because methane is more than 28x more potent than CO₂ over a 20-year horizon.³ These benefits are in addition to other costs that may be avoided depending on the regulatory environment, and improvements in overall operational well efficiency gained by making use of associated gas.

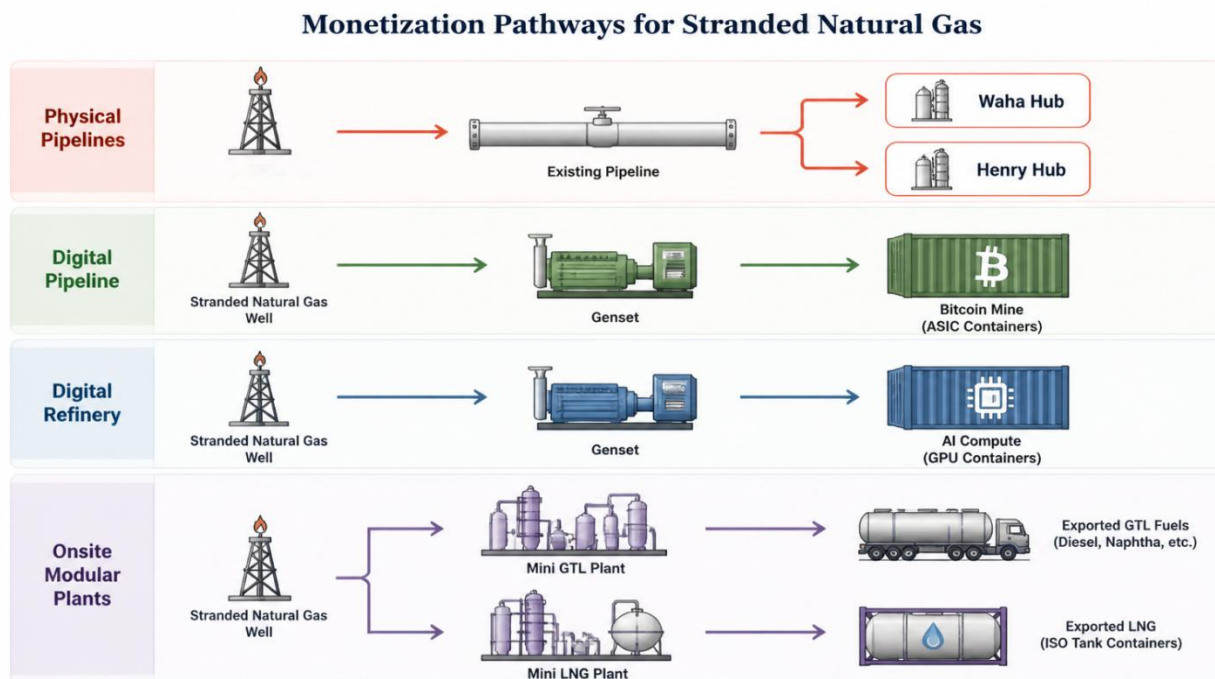
The report concludes with a strategic comparison of each pathway, suggesting that bitcoin mining offers the best natural gas use for sites that are isolated and unable to take advantage of other options, while AI compute is by far the most lucrative but requires certain conditions such as higher connectivity and concentration of natural gas. The Mini-GTL and Mini-LNG are great midstream options that offer more revenue than traditional pipelines and mining, and provide valuable end products that can be used in a global market.

Introduction

Overview of Monetization Pathways

In the context of the upstream oil and gas industry, natural gas produced as a byproduct of crude oil extraction is often geographically isolated from pipeline infrastructure. Even when pipeline infrastructure is available, a surplus of local supply, combined with takeaway constraints, can lead to suppressed, even negative prices.⁴ When the cost of constructing

takeaway capacity outweighs the value of the gas, operators have traditionally resorted to routine flaring or venting. Rather than leave energy (and consequently, money) on the table, there are more options than ever before to monetize this stranded gas to cover a variety of different situations. There are additional options out there, but these are some of the most promising and exciting emerging technologies and solutions.



Digital Pipelines – Bitcoin Miners

We will start by looking at bitcoin mining. What makes the “digital pipeline” of bitcoin mining so compelling? It’s no longer about looking for a big payout (although under the right setup the revenue can be quite respectable, even outperforming hub prices). The real value proposition is that it’s a very scalable, portable, fast, and predictable solution that can act as a “first responder” to any wasted, stranded energy. Viewed this way, it is almost like the Starlink of energy monetization – better than nothing, and can go basically anywhere with internet service (with current Starlink coverage, that does mean just about *anywhere* in the world).⁵

This can be done with a minimal amount of gas processing and generation setup, and indeed there are companies, such as Upstream Data and EZ Blockchain that provide all-in-one solutions. The gas processing/genset, and mining datacenter are commonly provided in modular shipping container setups. A great overview and walkthrough of one of these setups can be seen below. Fundamentally, there are a couple of business models to consider here. One is for the well owner/operator to purchase a turnkey mining solution and capture all of the mining upside, as well as operational risk (this is Upstream Data’s model). I mention this in the

later analysis but did not go through all of the calculations to determine the equivalent FCI and payback period. The other quite interesting idea is for a mining company to essentially provide a revenue sharing deal with the well operator, and own and manage the mining rig (the model used by companies like EZ Blockchain and Gridless).⁶ This means that the operator does not need to invest any capital, making it truly like a virtual “pipeline” that they can sell gas into (either from mining revenue, or a comparable gas purchase deal). And while grid-based mining looks for the lowest possible electricity price, mining on stranded energy means that the miner will always have access to the lowest possible cost of energy – in some cases, almost zero.

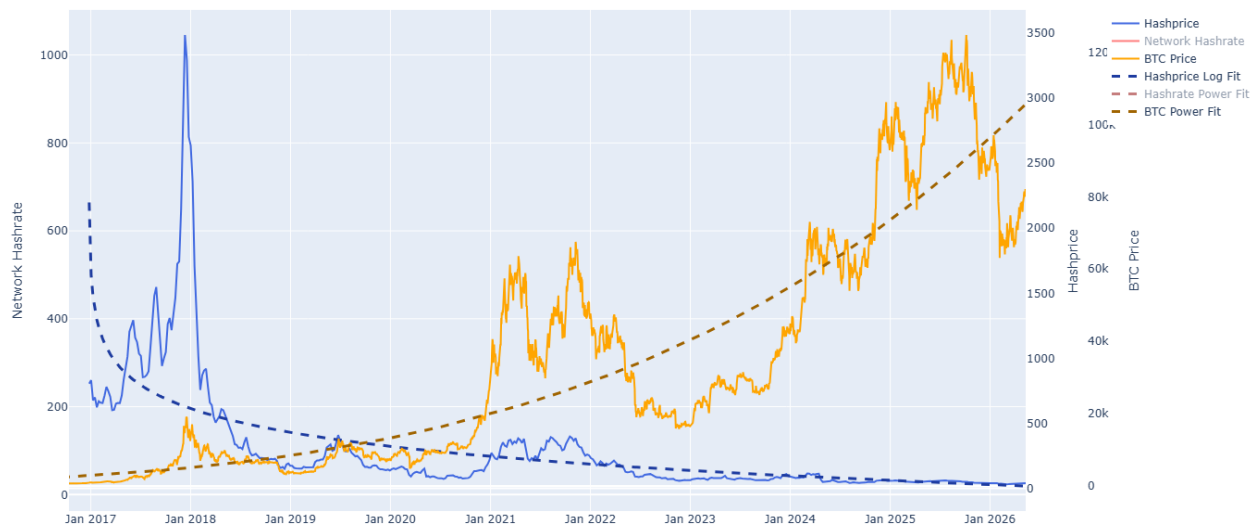


CREDIT: UPSTREAM DATA ([HTTPS://WWW.YOUTUBE.COM/WATCH?V=Q_1JRIZTVMQ](https://www.youtube.com/watch?v=Q_1JRIZTVMQ))

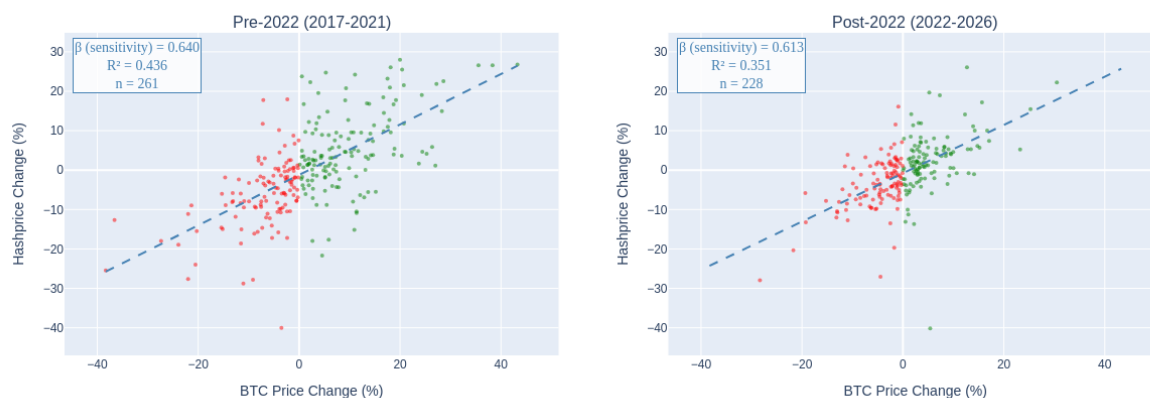
Although these mining operations are becoming a more structured industry, a common concern is the perceived uncertainty and volatility around bitcoin price. I will show why that should not be a major concern, now and into the future. Below are a couple of charts borrowed from another project I am working on. The top simply shows that bitcoin price best fits a power law model, while the hashprice log curve fits a smooth, monotonic decline. The diminishing effect that each bull market has had on hashprice, from 2018 to 2021-22, to 2025 is the key takeaway here. The second shows the percent change in hashprice vs percent change in bitcoin price over two different time periods, with decreasing sensitivity and correlation pointing to a decoupling between bitcoin price changes, and corresponding hashprice changes, as well as a tighter clustering of points suggesting a decrease in volatility.

The point of all this analysis is to make the case that, over time, the revenue stream from bitcoin mining activities appears likely to become more stable, and predictable. I felt confident enough in this conclusion to use simple trailing averages over different time periods in this report, rather than more complex price predictions. While there are still likely to be temporary events that cause hashprice shocks, the net effect of reduced volatility and the

continued growth in hashrate create a compelling case for improving revenue consistency. In addition, new forward contracts such as Luxor’s hashrate markets allow miners to hedge against uncertainty, further protecting profits against volatility.



Pre/Post 2022 Comparison: Hashprice Response to Bitcoin Price Changes (1 Week Intervals)



Digital Refinery – AI Compute

If bitcoin mining represents the truly location-agnostic energy consuming digital pipeline, the rapidly growing demand for power from AI compute “digital refineries”, and the similarities in their “powered shell” type of physical infrastructure, has sparked many companies’ interest in repositioning their energy resources towards providing firm supply to these datacenters. Any mining farms with the scale and infrastructure to support AI compute are rapidly being converted to service this kind of compute as the revenue potential (at least on paper) is many times higher than the revenue stream from mining. This makes perfect sense for regions with a high density of wells, and/or a more reliable high-bandwidth network connection that AI inferencing workloads require (although AI training workloads can also work

on remote systems with less connectivity). Companies like Crusoe and Coreweave have figured this out and are already racing to bring large amounts of power online in regions where the natural gas supply allows for it.⁷ While this analysis will briefly touch on what the economics of a hyperscale project might look like, the focus will be more on a newer, small modular class of AI compute since this is more easily comparable to the single well economics of other pathways. Similar to bitcoin mining containers, these products can be brought to a single small site, with some key differences that will be discussed later. Crusoe's recently announced Crusoe Spark product is at the right scale for a more natural comparison.



CREDIT: CRUSOE (<https://www.crusoe.ai/resources/newsroom/crusoe-announces-new-manufacturing-facility-to-produce-modular-ai-factories>)

Modularized GTL and LNG Providers

The final category to consider here are the new modular GTL and LNG technologies. Given the challenges discussed above with transporting gas through traditional pipelines, converting the natural gas onsite to high-value hydrocarbon liquids such as gasoline, diesel, and LNG is another great option as long as transportation and logistics networks are in place.⁸ Companies such as Galileo Technologies are developing impressive new technologies for liquefaction that can go onsite and remain cost-effective even at smaller volumes.⁹ These represent a longer-term, capital-intensive investment, but operationally they appear to be less risky and capital intensive than pursuing AI compute revenue. The estimates here are rough and do not perfectly capture the operational and equipment complexities, though they are based on existing research and give a good order of magnitude estimate to help understand where these options might fit in within the rest of the monetization pathways.

Techno-Economic Analysis (TEA)

All scenarios will consider a standard well size of 500 Mcf:

$$\begin{aligned}Q_{gas} &= 500 \text{ Mcf/day (single well)} \\Q_{annual} &= Q_{gas} \times 365 \text{ days/yr} = 182,500 \text{ Mcf/yr} \\E_{annual} &= Q_{annual} \times 1 \frac{\text{MMBtu}}{\text{Mcf}} = 182,500 \text{ MMBtu/yr}\end{aligned}$$

Physical pipelines will be assumed to have a high on-stream efficiency of 95%, while the digital solutions will assume a lower, but still optimistic efficiency of 85%.

Physical Pipelines

The pipeline calculations assume that an existing pipeline is already nearby and available to be sent for immediate sale. These represent a baseline to compare the other monetization options. Suppressed (negative) Waha, Henry, and a recent HH spike are used to capture a range of different possible pipeline revenue scenarios, with the on-stream efficiency of 0.95.

Waha Hub (Floor Price)

$$\begin{aligned}R_{Waha} &= 182,500 \text{ MMBtu/yr} \times (-\$8.00/\text{MMBtu}) \times 0.95 \\R_{Waha} &= -\$1,387,000/\text{yr}\end{aligned}$$

Henry Hub Current (EIA 4Q 2025 Forecast)

$$\begin{aligned}R_{HH,current} &= 182,500 \text{ MMBtu/yr} \times \$3.90/\text{MMBtu} \times 0.95 \\R_{HH,current} &= \$675,863/\text{yr}\end{aligned}$$

Henry Hub High (Extreme Scenario, Fall 2022 Peak)

$$\begin{aligned}R_{HH,high} &= 182,500 \text{ MMBtu/yr} \times \$8.00/\text{MMBtu} \times 0.95 \\R_{HH,high} &= \$1,387,000/\text{yr}\end{aligned}$$

Digital Monetization Pathways

Bitcoin Mining (Single Well) – Revenue Stream Service

The first digital monetization option to explore is the single well bitcoin mining setup. I used the hashprice-based methodology discussed above to explore a realistic outcome range. Additionally, the analysis will assume a revenue sharing approach where older hardware is commonly used, in this case an Antminer S19 XP — 141 TH/s @ 21.3 J/TH (3,000 W per unit). A PUE of 1.25 will be used as a reasonable assumption to an efficient air-cooled setup. Another assumption is that a minimal amount of gas conditioning/scrubbing, can be done onsite and is included in the cost. Additionally, the simple revenue share business model of 30% of gross

mining revenue will be used as the revenue “stream”, meaning that there is no fixed capital investment, or FCI. The other option of procuring all of the mining equipment would typically result in a ~2-3-year payback period, and ultimately a higher revenue (as seen without the 30% scaling).

Step 1: Power Output (Small Engine/Turbine, HR = 10,000 Btu/kWh)

$$P_{well} = \frac{500 \times 10^6}{10,000 \times 24} = 2.083 \text{ MW} = 2,083 \text{ kW}$$

Step 1b: Apply PUE Derating (PUE = 1.25)

$$P_{IT} = \frac{P_{well,small}}{1.25} = \frac{2.083}{1.25} = 1.667 \text{ MW}$$

$$P_{IT,available} = 1.667 \times 10^6 = 1,666,667 \text{ W}$$

Step 2: Miner Power and Count (\$19 XP)

$$P_{miner} = 3,000 \text{ W}, N_{miners} = \left\lfloor \frac{1,666,667}{3,000} \right\rfloor = 555 \text{ miners}$$

Step 3: Fleet Hashrate

$$H_{fleet} = 555 \times 141 \text{ TH/s} = 78,255 \text{ TH/s} = 78.3 \text{ PH/s}$$

Step 4: Apply Hashprice (30-day avg, Apr 12-May 11 '26)

$$R_{gross} = 78.3 \text{ PH/s} \times \$35.50/\text{PH/s/day} \times 365 = \$1,013,989/\text{yr}$$

Step 5: Apply On-Stream Efficiency (85%)

$$R_{eff} = R_{gross} \times \eta = \$1,013,989 \times 0.85 = \$861,891/\text{yr}$$

Step 6: Apply Operator Revenue Share (30%)

$$R_{operator} = R_{eff} \times f_{share} = \$861,891 \times 0.30 = \$258,567/\text{yr}$$

Step 7: Find Equivalent gas value:

$$P_{equiv} = \frac{R_{operator}}{E_{annual}} = \frac{\$258,567}{182,500} \approx \$1.42/\text{MMBtu}$$

For the high end of the range, I will use the post-halving average (Apr '24-May '26):

Hashprice (\$/PH/s/day)	Gross Revenue	Operator Revenue (30%)	Equiv. \$/MMBtu
\$48.21	~\$1,376,044/yr	~\$351,142/yr	~\$1.92/MMBtu

So, the expected annual revenue an operator could receive might range between ~\$250k-350k a year, or a MMBtu of ~\$1.42-1.92, which is a little less than typical Henry Hub prices. That is pure revenue, with minimal effort on the operator's part, from gas that would be flared!

Modular AI Compute (Single Well) – Crusoe “Spark” Model

The second option, AI compute, is by far the most lucrative monetization pathway even using simplified assumptions. The need for “speed to power” driven by the current insatiable demand of new AI datacenters is a major energy monetization driver that could deliver orders of magnitude more net profit per unit of gas consumed. It also comes with its own set of risks and challenges that should be considered, which will be discussed in the results.

For the AI compute analysis, it was necessary to do a ground-up capital expenditure approach, similar to the modular conversion plants. I would have liked to have a way to compare it with a revenue sharing type of business model like the Bitcoin setup described above, but there are a couple of issues. The first is that there don’t seem to be any revenue sharing AI compute-type services yet (Crusoe Spark Edge Zones might be the closest thing, but they are vertically integrated). The second is that the revenue “stream” isn’t as transparent and straightforward to calculate. So, I attempted to roughly estimate revenue by following Crusoe’s model (they do acquire gas from wells, but at a price of 10-30 cents per Mcf – at which level the comparison would miss the upside opportunity available to natural gas producers).¹⁰

The first model is a single-well, modular/edge AI compute setup – similar in size to the bitcoin mine above. The same PUE of 1.25 appears to be reasonable and makes for an even comparison. Let’s start by using the same calculated power per well as the small engine bitcoin mining setup:

Step 1: Power Output (Small Engine/Turbine)

$$P_{well} = 2.083 \text{ MW}$$

Step 1b: Apply PUE Derating (PUE = 1.25)

$$P_{IT} = \frac{P_{well}}{1.25} = \frac{2.083}{1.25} = 1.667 \text{ MW}$$

Step 2: GPU Fleet Sizing

Assumes NVIDIA H100 clusters @ ~700W–1kW per node, including networking/cooling

$$N_{GPU} = \lfloor P_{well} \times \rho_{GPU} \rfloor = \lfloor 1.667 \times 1,000 \rfloor = 1,667 \text{ GPUs}$$

Step 3: Fixed Capital Investment (Spark)

I estimated \$1,500 per kW for all-in generation and associated power costs.¹¹ Also, \$12,000 per kW for a rough estimate of construction costs, from a recent 2026 Global Data Center Outlook report, as well as an assumed cost of \$30,000 per H100 GPU¹²:

$$FCI_{power} = c_{power} \times P_{kW} = \$1,500/\text{kW} \times 2,083 \text{ kW} = \$3,125,000$$

$$FCI_{shell} = c_{shell,spark} \times P_{kW} = \$12,000/\text{kW} \times 2,083 \text{ kW} = \$25,000,000$$

$$FCI_{IT} = N_{GPU} \times c_{GPU} = 1,666 \times \$30,000 = \$49,980,000$$

$$FCI_{total} = FCI_{power} + FCI_{shell} + FCI_{IT} = \$78,105,000$$

Step 4: Annual Gross Sales

Crusoe publishes on-demand pricing for H100 as well as many other models.¹³ A conservative estimate of \$2.80/hr is used, with a best-case assumption of 85% annual utilization, to match with the assumed on-stream efficiency value for the other processes:

$$S = N_{GPU} \times R_{hr} \times H_{yr} \times \eta$$
$$S = 1,667 \times \$2.80/\text{hr} \times 8,760 \text{ hr} \times 0.85 = \$34,734,101/\text{yr}$$

Step 5: Net Profit

Next, we will assume an OpEx value of 20% of gross sales to cover engineering staff, platform maintenance, and other remote field operations, and get the annual net profit:

$$\text{OpEx}_{\text{spark}} = 0.20 \times S = \$6,946,820/\text{yr}$$
$$\pi_{\text{net}} = S \times (1 - 0.20) = \$34,734,101 \times 0.80 = \$27,787,281/\text{yr}$$

Step 6: Simple Payback Period

$$PBP = \frac{\text{FCI}_{\text{total}}}{\pi_{\text{net}}} = \frac{\$78,105,000}{\$27,787,281} = 2.81 \text{ yr}$$

Step 7: Equivalent Gas Value

$$P_{\text{equiv}} = \frac{\pi_{\text{net}}}{E_{\text{annual}}} = \frac{\$27,787,281}{182,500 \text{ MMBtu}} \approx \$152.26/\text{MMBtu}$$

Hyperscale AI Compute (Multi-Well) – Crusoe “Stargate”

This is essentially a scale-up to a 100MW reference campus which would include a large gas turbine (HR = 8,000 Btu/kWh) for scale-appropriate power generation. Scale economies could reduce the DC shell cost per kW and the OpEx ratio, but the other details are mostly the same. Running the same kind of analysis reveals even more net profit per year (\$43,428,050/yr), also with a shorter payback period (~2.26 years) and an equivalent gas value of ~\$ 202/MMBtu. Because this is such a large, and complex undertaking, I felt that the analysis here may be too far off and not as comparable to the scale and scope of the other monetization projects in this report, so I will not go into the full details here.

Modular GTL and LNG Conversion

Finally, we can compare the above options with a mini-GTL plant, using the turnover ratio method to estimate revenue. The FCI values used are from Ravindran and El-Halwagi (2023) for small to mid-size GTL processing units.⁸

Step 1: Convert Gas to Product Capacity

Conversion factor: ~10 Mcf gas -> 1 bbl GTL product

$$C_{\text{GTL}} = \frac{Q_{\text{gas}}}{10} = \frac{500 \text{ Mcf/day}}{10} = 50 \text{ bbl/day}$$

Step 2: Estimate FCI Using Correlation

From Ravindran and El-Halwagi (2023), valid for 10-15,500 bbl/day:

$$\begin{aligned} FCI_{GTL} &= 0.12 \times C^{0.96} \\ FCI_{GTL} &= 0.12 \times (50)^{0.96} \\ FCI_{GTL} &= 0.12 \times 42.77 = \$5.13 \text{ MM} \end{aligned}$$

Step 3: Estimate Revenue Using Turnover Ratio

Turnover ratio for GTL $\approx$ 0.5:

$$\begin{aligned} R_{GTL} &= FCI_{GTL} \times \text{Turnover Ratio} \\ R_{GTL} &= \$5.13 \text{ MM} \times 0.5 = \$2.57 \text{ MM/yr} \end{aligned}$$

Step 4: Calculate Simple Payback Period

$$PBP = \frac{FCI}{R_{\text{annual}}} = \frac{\$5.13 \text{ MM}}{\$2.57 \text{ MM/yr}} = 2.0 \text{ years}$$

Using a similar calculation process, Mini-LNG was also explored, and found to have a revenue of $\sim$ \$9.87 MM/yr over roughly the same 2-year payback period. This is a very simple PBP estimate and does not account for taxes and depreciation.

Results and Discussions

The final pathway comparison results are summarized in the table below:

Pathway	FCI	Annual Revenue	Equiv. \$/MMBtu ↓
Pipeline — Waha (Floor)	\$0	(\$1.39 MM)	(\$8.00)
Bitcoin Mining (Service)	\$0	$\sim$ \$0.26-\$0.35 MM	$\sim$ \$1.42-\$1.92
Pipeline — HH (Current)	\$0	\$0.68 MM	\$3.90
Bitcoin Mining (Owned)	<TBD>	$\sim$ \$0.86-\$1.37 MM	$\sim$ \$4.72-\$7.53
Pipeline — HH High	\$0	\$1.39 MM	\$8.00
Mini-GTL	$\sim$ \$5.13 MM	$\sim$ \$2.57 MM	$\sim$ \$14.06
Mini-LNG	$\sim$ \$19.75 MM	$\sim$ \$9.87 MM	$\sim$ \$54.10
Crusoe Spark (per well)	$\sim$ \$78.1 MM	$\sim$ \$27.8 MM	$\sim$ \$152

From these results, even though there are a lot of assumptions baked in, we can see that the revenue ranges fall into sensible ranges for each monetization option. The revenue multiple going from bitcoin (owned) to AI compute is actually quite close to one industry reported value of a 15x increase.¹⁴ At the bottom layer would be revenue sharing type bitcoin mining operations, or “better than nothing” monetization option in highly remote, inaccessible locations where there is no alternative (or, pipeline revenue is negative due to oversupply, as in

the case of Waha). A company owning and optimizing their own mining operations could optimistically earn closer to ~\$1MM per year, bringing the revenue much closer to typical pipeline prices. An opportunistic gas owner could even have a physical pipeline as well as a mining operation, switching to whichever process the market economics favor most at a given time, and one company has been using exactly this model.¹⁵ Physical pipelines operations when prices are high can exceed mining revenue, although this is highly market dependent. The modular conversion plants occupy the next tier, sitting between pipeline revenue “streams” and on-site power generation, bringing in anywhere from ~\$2-10MM annually. At the highest tier, building and operating the entire vertical of AI compute as in Crusoe’s case, represents an extraordinarily outsized annual revenue opportunity (although this would be extremely capital and operationally intensive to pull off).

There are a couple of missing “hidden costs” that were not considered for the digital services in the simple models above. The first is that a minimal gas conditioning unit would be needed to dehydrate and scrub H₂S if dealing with sour gas. The other, and potentially larger one is the “refresh cycle” wall – both AI GPUs and certain classes of bitcoin mining machines can become functionally obsolete in 3-5 years. There are strategies for mitigating this for both types of compute, but it’s a considerable operational cost that both mining companies and AI compute providers will have to take into account. By comparison, the Mini GTL and LNG plants can offer a longer term “set it and forget it” infrastructure once built.

Using stranded natural gas specifically for AI compute as a service has a number of challenges and tradeoffs that must be considered alongside the incredible upside potential. Although even at the small scale it looks operationally very similar to a bitcoin mine (natural gas onsite processing and generation, and a mini-datacenter compute farm with PDUs and cooling), there are some key differences. The first is operational flexibility – AI compute centers like the Spark are constrained by service quality and workload structure in ways that bitcoin mining is not. AI workloads are latency sensitive when used for inference (though these can also be used for training in more remote areas), meaning they are more geographically than miners. Furthermore, they are often tied to SLAs and user experience, meaning that excessive downtime must be avoided, and additional costs like battery backup will probably need to be considered for these deployments. Finally, the monetization or revenue stream is dependent on users paying for compute time – companies like Crusoe have pricing that can change based on demand and impact bottom line revenue.¹³ This means that although demand is high (and could indeed stay high given continued AI inference demand), revenue is determined in a way that likely makes it harder to offer revenue sharing for an energy “stream” in the same way as bitcoin mining. In conclusion, AI compute is the high risk, high reward option, but currently, it is more like running a highly technical startup on the back of an oil well. In the future this will likely continue to evolve and more competition may allow for new ways to do AI compute.

Conclusion

The use of one monetization pathway or another is not a one-size-fits-all situation – the answer of which technology to use, as the saying goes is: “it depends”. Throughout this report I have been especially interested in reviewing the digital opportunities – one, bitcoin mining, being a more slow and steady revenue stream with the highest degree of operational flexibility: it can be set up at just about any site, no process restrictions, and (in the revenue sharing case), no upfront capital costs. And the other, AI compute, being the higher-risk, high reward alternative that can look similar infrastructure-wise at a smaller scale, but is also much more capital intensive. Hopefully it is clear that any well with gas flow and limited or no pipeline access should at the very least seriously consider owning or partnering with a bitcoin mining company as a potential economic upgrade. The modular Mini-GTL and LNG plant options may be best for operators who have existing midstream expertise or partnerships and are comfortable deploying the capital, and managing the supply chain logistics inherent with each process. The takeaway here seems to be that both physical and the bitcoin digital pipeline pathways offer straightforward revenue streams, but investing the capital to build any of the other solutions offers much higher returns on natural gas in the long run. This also does of course require more operational overhead and industry expertise. One final factor to mention is that a large amount of takeaway capacity may be coming online for the Waha hub soon, and it will be interesting to see how it affects traditional pipeline pricing, and how it impacts the stranded gas situation.⁴ Any of these solutions offer a much better use for stranded natural gas than flaring, and with so many great solutions in play, hopefully we will continue to see less and less waste in the natural gas sector.

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